

Problem Set 1

Due on Canvas at noon Friday September 18th

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 1 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions.** Grades will be finalized a few days after the due date.

Material covered: This problem set covers the first six lectures (up to Sept 17th)

Problem 1. Boycotts

Note: This problem has been a tradition at Yale SOM for many years. It is a rite of passage for SOM economics students. To answer this problem requires no mathematics beyond arithmetic. Read the instructions carefully, and then feel free to talk about this problem with your fellow students. We advise such discussions to prevent you from wasting time. In the end you must see the answer yourself in your own “aha” moment and of course, the written answer must be your own.

Setup A frequently used instrument of economic coercion is the boycott. For example, a union involved in a serious and protracted conflict with some employer or group of employers may urge all of its sympathizers to stop buying the products of that employer or group, in the hope that the resulting economic losses will provide an incentive for the employers to make concessions. However, experience shows that the coercive power of a boycott is often small.

In this exercise, you are to use supply and demand analysis as a tool to understand the effects of boycotts. For the sake of concreteness and precision, most of the questions are posed in quantitative terms. However, the goal of the exercise is to develop your understanding of the logic of the market processes involved. Accordingly, if you have difficulty, don't try to overwhelm the problem with calculation -- rethink the economics.

Throughout the exercise, the following assumptions are to be maintained. First, the market in question is one to which simple supply and demand analysis is applicable: there are numerous buyers and sellers in the market; the good is homogeneous; buyers and sellers seek only to trade on the most advantageous possible terms and, aside from the complications introduced by the boycott, do not care with whom they transact. Second, the identities of buyers and sellers are fixed: in particular, no reselling of the goods takes place. Third, the set of buyers divides sharply into observers and nonobservers of the boycott, while the sellers separate equally sharply into those that are targets of the boycott and those that are not. Boycott observers will not, under any circumstances, buy from a target seller, while non-observers do not care from whom they buy. For simplicity, it is assumed that boycott observers account for the same fraction of demand at every price. Similarly, target sellers account for a certain fraction of the supply at every price.

Assume initially that in the pre-boycott situation, the market demand curve is described by:

$$Q_D = 5000 - 10P$$

where Q_D = quantity demanded and P = the price. The market supply curve has equation (with Q_S = quantity supplied):

$$Q_S = 50(P - 5)$$

except that, at prices of 5 or less, quantity supplied is zero.

Warmup problem (i): Determine the market equilibrium in the pre-boycott situation.

Answer: In equilibrium, the price is such that quantity demanded equals quantity supplied:

$$\begin{aligned}Q_D &= Q_S \\5000 - 10P &= 50(P - 5) \\60P &= 5250 \\P &= 87.5 \\Q &= 4125\end{aligned}$$

Warmup problem (ii): Suppose that buyers representing 70% of the original demand, observe the boycott. If all non-observers buy from target sellers, what is the demand curve facing target sellers?

Answer: Since observers will not buy from target sellers at any price, the only demand facing target sellers is from non-observers. Since observers and non-observers account for demand in the ratio 0.7:0.3 at every price, the equation of the demand curve facing target sellers is:

$$Q_D = 0.3(5000 - 10P) = 1500 - 3P$$

Questions

Consider the three situations characterized by the following table

<u>Situation</u>	<u>Observers as Fraction of Original Demand</u>	<u>Target as Fraction of Original Supply</u>
A	3/4	3/4
B	1/4	1/4
C	1/2	1/4

For each of the three situations, determine the market equilibrium that would arise under boycott conditions. Give the quantities transacted by observing and non-observing buyers, and by target and non-target sellers.

Drawing on your results, state the general principle which relates the effectiveness of a boycott to the observance fraction and the target fraction. "Effectiveness," in this context, means lowering the price to target sellers. Can you provide some intuition as well?

Hint: Warmup problem (i) supplies all the mathematical knowledge you will need to solve the remaining problems. What is needed for the rest is an ability to structure the problem, and an understanding of simple economic principles. The point of the exercise is to determine whether the boycott is effective in any of the three cases above. To answer this question, you have been given

the following information: (1) the market supply curve, (2) the targeted fraction of sellers, (3) the market demand curve, (4) the observers as a fraction of buyers. Items (1) and (2) imply that it is possible to write down algebraic expressions for the supply curves of targeted and non-targeted sellers. You can also choose to use Excel to get a sense of how prices get determined in each sub-market.

Problem 2. Black quinoa in Bolivia

The Government of Bolivia is planning to impose a price control in the market for black quinoa, an important part of the standard diet of people in the country. Market demand for black quinoa is given by $Q_d = 40 - P$ and the market supply curve is given by $Q_s = -20 + 3P$, where P denotes the market price in dollars per kilogram and Q_d and Q_s denote the quantities demanded and supplied, respectively (measured in millions of kilograms per year).

- a. What is the equilibrium market price and quantity? What is the consumer surplus and the producer surplus?

a. Answer: $40 - P = -20 + 3P$

$$P = 15$$

$$\text{So, } Q = 25$$

$$C.S. = (1/2) * 25 * 25 = 312.25$$

$$P.S. = (1/2) * 25 * (25/3) = 104.17$$

- b. Find the elasticity of demand at the equilibrium price? Is the demand for quinoa elastic or inelastic? Is this elasticity consistent with what you expected? Explain

a. Answer: Elasticity = $-15/25 = -0.6$

So demand is inelastic which is consistent with quinoa being an essential part of the Bolivian diet.

- c. Consider the situation where the Government imposes a price control limiting market price to no higher than \$10. What is the new equilibrium price and quantity? What are the resulting consumer surplus, producer surplus and total surplus?

a. Answer: New price is \$10

$$\text{New quantity is } 10$$

$$C.S. = (1/2) * 10 * 10 + 20 * 10 = 250$$

$$P.S. = (1/2) * 10 * (10/3) = 16.67$$

$$\text{Total Surplus} = 266.67$$

- d. Looking at the effect of the price control on the total surplus, what would you say? Is this an efficient step? Why or why not?
 - a. Answer: This was not an efficient step since total social surplus went down.

- e. What if the government limited the market price to \$15.50? Would this be a smart move? Why or why not?
 - a. Answer: A price control of \$15.50 would not have any effect since the equilibrium price is already below \$15.50.